

UNIT III

OBJECTIVE:

To familiarize the students with the topic of exergy and to learn about the practical applications of second law efficiencies

DEAD STATE:

A system exchanges work, heat and mass with its surroundings during the process. If the system reaches a state which is in equilibrium with its surrounding then the system cannot exchange the work heat and mass with its surroundings. This state is called a dead state and its properties are denoted by subscript O, such as pressure P_o and temperature T_o .

At the dead state:

- A system is at the temperature and pressure of its surroundings.
- It has no kinetic (or) potential energy relative to its surroundings
- It does not react with the surroundings
- There are no unbalanced magnetic, electrical and surface tension effects b/w the system and its surroundings.

AVAILABILITY:

The availability of a given system is defined as the maximum useful work (total work $-PdV$ work) that is obtainable in a process in which the system comes to equilibrium with its surrounding availability is thus a composite property depending on the state of system and surroundings.

Whenever useful work is obtained during a process in which a finite system undergoes a change of state, the process must terminate.

Therefore for a closed system availability can be expressed as

$$\phi = (U - U_o) + P_o (V - V_o) - T_o (S - S_o)$$

Similarly for an open system

$$\psi = (H - H_o) - T_o (S - S_o)$$

In steady flow system the exit condition is assumed to be in equilibrium with the surroundings. The change in availability of a system when it moves from one state to another can be given as for closed system

$$\phi_1 - \phi_2 = (U_1 - U_2) + P_o (V_1 - V_2) - T_o (S_1 - S_2)$$

For an open system

$$\psi_1 - \psi_2 = (H_1 - H_2) - T_o (S_1 - S_2)$$

IRREVERSIBILITY:

In thermodynamics, a change in the thermodynamic state of a system and all of its surroundings cannot be precisely restored to its initial state by infinitesimal changes in some property of the system without expenditure of energy. A system that undergoes an irreversible process may still be capable of returning to its initial state; however, the impossibility occurs in restoring the environment to its own initial conditions. An irreversible process increases the entropy of the universe. However, because entropy is a state function, the change in entropy of the system is the same whether the process is reversible or irreversible. The second law of thermodynamics can be used to determine whether a process is reversible or not

The actual work done by a system is always less than the idealized reversible work, and the difference between the two is called the irreversibility of the process.

$$I = W_{\max} - W$$

This is also some time referred to as degradate or dissipation. For a non-flow process between the equilibrium states, when the system exchanges heat only with the environment.

$$\begin{aligned} I &= [(U_1 - U_2) - T_0(S_1 - S_2)] - [(U_1 - U_2) + Q] \\ &= [T_0(S_2 - S_1) - Q] \\ &= T_0(\Delta S)_{\text{system}} + T_0(\Delta S)_{\text{surrounding}} \\ &= T_0[(\Delta S)_{\text{system}} + T_0(\Delta S)_{\text{surrounding}}] = T_0(\Delta S)_{\text{universe}} \end{aligned}$$

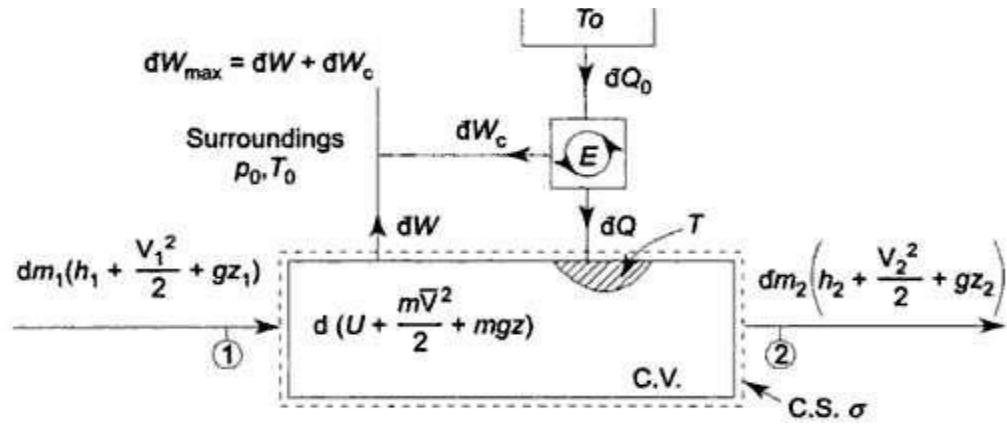
Similarly, for the steady flow process

$$\begin{aligned} I &= W_{\max} - W \\ &= T_0(S_2 - S_1) - Q \\ &= T_0(\Delta S)_{\text{system}} + T_0(\Delta S)_{\text{surrounding}} \\ &= T_0[(\Delta S)_{\text{system}} + T_0(\Delta S)_{\text{surrounding}}] = T_0(\Delta S)_{\text{universe}} \end{aligned}$$

The same expression for irreversibility applies to both flow & non – flow process.

REVERSIBLE FOR IN STEADY FLOW PROCESS:

Let us consider and open system exchange energy only with the surrounding at constant temperature T_0 and at constant pressure P_0 . A mass d_{m1} enters the system at state 1, a mass d_{m2} leaves the system at state 2, an amount of heat dQ is absorbed by the system an amount of work dw is delivered by the system, and the energy of the system changes by an amount $d(E)$ applying the first law, we have,



For a steady flow process

$$dm_1 = dm_2 = dm$$

and

$$d \left[U - T_0 S + \frac{mV^2}{2} + mgz \right]_{\sigma} = 0$$

Equation (8.24) reduces to

$$dW_{\max} = dm \left[\left(h_1 - T_0 s_1 + \frac{V_1^2}{2} + gz_1 \right) - \left(h_2 - T_0 s_2 + \frac{V_2^2}{2} + gz_2 \right) \right]$$

For total mass flow, the integral form of Eq. (8.25) becomes

$$W_{\max} = \left(H_1 - T_0 S_1 + \frac{mV_1^2}{2} + mgz_1 \right) - \left(H_2 - T_0 S_2 + \frac{mV_2^2}{2} + mgz_2 \right)$$

The expression $(H - T_0 S)$ is called the *Keenan function*, B .

$$\therefore W_{\max} = \left(B_1 + \frac{mV_1^2}{2} + mgz_1 \right) - \left(B_2 + \frac{mV_2^2}{2} + mgz_2 \right)$$

$$= \psi_1 - \psi_2$$

where ψ is called the *availability function* of a steady flow process given by

$$\psi = B + \frac{mV^2}{2} + mgz$$

On a unit mass basis,

$$W_{\max} = \left(h_1 - T_0 s_1 + \frac{V_1^2}{2} + gz_1 \right) - \left(h_2 - T_0 s_2 + \frac{V_2^2}{2} + gz_2 \right)$$

$$= \left(b_1 + \frac{V_1^2}{2} + gz_1 \right) - \left(b_2 + \frac{V_2^2}{2} + gz_2 \right)$$

If K.E. and P.E. changes are neglected

$$W_{\max} = B_1 - B_2$$

$$= (H_1 - T_0 S_1) - (H_2 - T_0 S_2)$$

$$= (H_1 - H_2) - T_0(S_1 - S_2)$$

REVERSIBLE WORK IN A CLOSED SYSTEM:

$$dm_1 = dm_2 = 0$$

$$\begin{aligned} dW_{\max} &= -d \left[U - T_0 S + \frac{m V^2}{2} + mgz \right]_{\sigma} \\ &= -d(E - T_0 S)_{\sigma} \end{aligned}$$

where

$$E = U + \frac{m V^2}{2} + mgz$$

For a change of state of the system from the initial state 1 to the final state 2,

$$\begin{aligned} W_{\max} &= E_1 - E_2 - T_0 (S_1 - S_2) \\ &= (E_1 - T_0 S_1) - (E_2 - T_0 S_2) \end{aligned}$$

If the K.E. and P.E. changes are neglected, Eq. (8.31) reduces to

$$W_{\max} = (U_1 - T_0 S_1) - (U_2 - T_0 S_2)$$

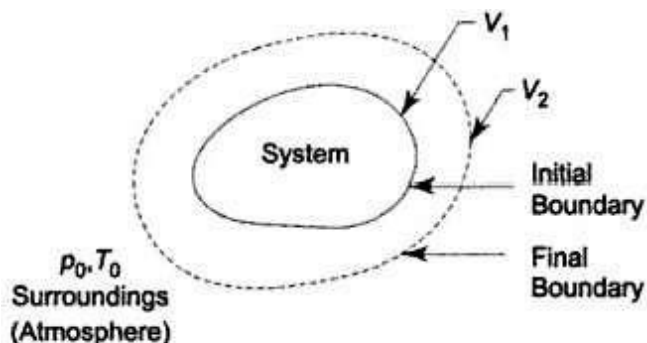
For unit mass of fluid,

$$\begin{aligned} W_{\max} &= (u_1 - u_2) - T_0 (s_1 - s_2) \\ &= (u_1 - T_0 s_1) - (u_2 - T_0 s_2) \end{aligned}$$

USEFUL WORK:

All of the work w of the system would not be available for delivery, since a certain portion of it would be spent in pushing out the atmosphere. The useful work is defined as the actual work delivered by a system less the work performed on the atmosphere if V_1 and V_2 are the initial and final volumes of the system and P_0 is the atmospheric pressure, then the work done on the atmosphere is $P_0 (V_2 - V_1)$. therefore, the useful work W_u becomes

$$W_u = W_{\text{act}} - P_0 (V_2 - V_1)$$



Work done by a closed system in pushing out the atmosphere

Similarly, the *maximum useful work* will be

$$(W_u)_{\max} = W_{\max} - p_0 (V_2 - V_1)$$

In differential form

$$(dW_u)_{\max} = dW_{\max} - p_0 dV$$

In a steady flow system, the volume of the system does not change. Hence, the maximum useful work would remain the same, i.e., *no work is done on the atmosphere*, or

$$(\delta W_u)_{\max} = dW_{\max}$$

But in the case of an unsteady-flow open system or a closed system, the volume of the system changes. Hence, when a system exchanges heat only with the atmosphere, the maximum useful work becomes

$$(\delta W_u)_{\max} = \delta W_{\max} - p_0 dV$$

Substituting $dW_{\max}$ from Eq.

$$\begin{aligned} (\delta W_u)_{\max} = & dm_1 \left(h_1 - T_0 s_1 + \frac{V_1^2}{2} + gz_1 \right) - dm_2 \left(h_2 - T_0 s_2 + \frac{V_2^2}{2} + gz_2 \right) \\ & - d \left[U + p_0 V - T_0 S + \frac{m V^2}{2} + mgz \right]_{\sigma} \end{aligned}$$

This is the maximum useful work for an *unsteady open system*.

For the closed system

$$\begin{aligned} (dW_u)_{\max} &= -d \left[U + p_0 V - T_0 S + \frac{m V^2}{2} + mgz \right]_{\sigma} \\ &= -d [E + p_0 V - T_0 S]_{\sigma} \end{aligned}$$

$$\therefore (W_u)_{\max} = E_1 - E_2 + p_0 (V_1 - V_2) - T_0 (S_1 - S_2)$$

If K.E. and P.E. changes are neglected

$$(W_u)_{\max} = U_1 - U_2 + p_0 (V_1 - V_2) - T_0 (S_1 - S_2)$$

This can also be written in the following form

$$\begin{aligned} (W_u)_{\max} &= (U_1 + p_0 V_1 - T_0 S_1) - (U_2 + p_0 V_2 - T_0 S_2) \\ &= \phi_1 - \phi_2 \end{aligned}$$

where ϕ is called the *availability function for a closed system* given by

$$\phi = U + p_0 V - T_0 S$$

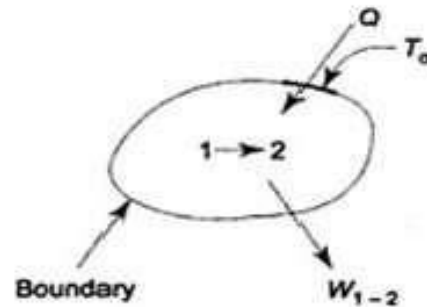
The useful work per unit mass becomes

$$(W_u)_{\max} = (u_1 + p_0 v_1 - T_0 s_1) - (u_2 + p_0 v_2 - T_0 s_2)$$

EXERGY BALANCE OF CLOSED AND OPEN SYSTEM:

CLOSED SYSTEM:

For a closed system, availability or exergy transfer occurs through heat and work interactions



Exergy balance for a closed system

1st law:
$$E_2 - E_1 = \int_1^2 \delta Q - W_{1-2}$$

2nd law:
$$S_2 - S_1 - \int_1^2 \left[\frac{\delta Q}{T} \right] = S_{\text{gen}}$$

or
$$T_0(S_2 - S_1) - T_0 \int_1^2 \left[\frac{\delta Q}{T} \right]_{\sigma} = T_0 S_{\text{gen}}$$

Subtracting

$$\begin{aligned} E_2 - E_1 - T_0(S_2 - S_1) &= \int_1^2 \delta Q - W_{1-2} - T_0 \int_1^2 \left[\frac{\delta Q}{T} \right]_{\sigma} - T_0 S_{\text{gen}} \\ &= \int_1^2 \left[1 - \frac{T_0}{T_{\sigma}} \right] \delta Q - W_{1-2} - T_0 S_{\text{gen}} \end{aligned}$$

Since,

$$A_2 - A_1 = E_2 - E_1 + p_0(V_2 - V_1) - T_0(S_2 - S_1)$$

$$A_2 - A_1 = \int_1^2 \left(1 - \frac{T_0}{T_{\sigma}} \right) \delta Q - [W_{1-2} - p_0(V_2 - V_1)] - T_0 S_{\text{gen}}$$

Change in exergy	Exergy transfer with heat	Exergy transfer with work	Exergy destruction
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OPENSYSYTEM:

1st law:

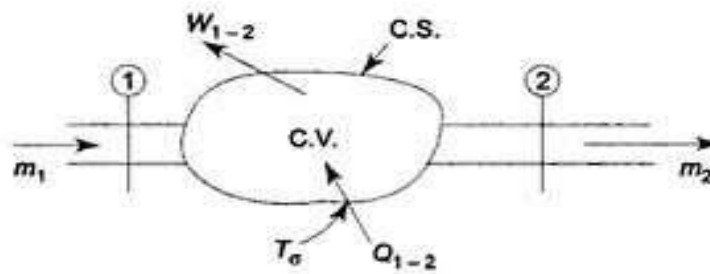
$$H_1 + \frac{m V_1^2}{2} + mgZ_1 + Q_{1-2} = H_2 + \frac{m V_2^2}{2} + mgZ_2 + W_{1-2}$$

2nd law:

$$S_1 + \int_1^2 \left(\frac{dQ}{T} \right) - S_2 = S_{\text{gen}}$$

or

$$T_0(S_1 - S_2) + T_0 \int_1^2 \left(\frac{dQ}{T} \right) = T_0 S_{\text{gen}} = I$$



Exergy balance for a steady flow system

From Eqs

$$H_2 - H_1 - T_0(S_2 - S_1) + m \frac{V_2^2 - V_1^2}{2} + mg(Z_2 - Z_1) = \int_1^2 \left(1 - \frac{T_0}{T} \right) dQ - W_{1-2} - I$$

or

$$A_2 - A_1 = \int_1^2 \left(1 - \frac{T_0}{T} \right) dQ - W_{1-2} - I$$

In the form of rate equation at steady state:

$$\sum_j \left(1 - \frac{T_0}{T_j} \right) \dot{Q}_j - \dot{W}_{\text{C.V.}} + \dot{m}(a_{f_1} - a_{f_2}) - \dot{I}_{\text{C.V.}} = 0$$

where $a_{f_1} - a_{f_2} = (h_1 - h_2) - T_0(s_1 - s_2) + \frac{V_1^2 - V_2^2}{2} + g(Z_1 - Z_2)$ and $[1 - T_0/T_j] \dot{Q}_j$

= time rate of exergy transfer along with heat $\dot{Q}_j$ occurring at the location on the boundary where the instantaneous temperature is T_j .

For a single stream entering and leaving, the exergy balance gives

$$\left[1 - \frac{T_0}{T_\sigma} \right] \frac{\dot{Q}}{\dot{m}} + a_{f_1} - \frac{\dot{W}}{\dot{m}} - a_{f_2} = \frac{\dot{I}}{\dot{m}}$$

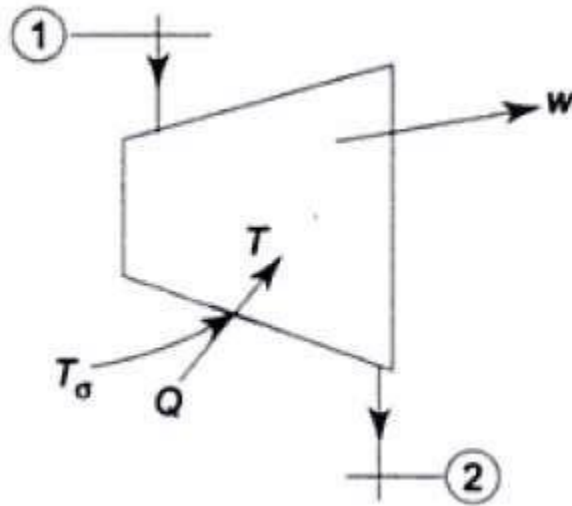
Exergy in Exergy out Exergy loss

SECOND LAW EFFICIENCIES OF THERMAL EQUIPMENTS:

1. TURBINE:

The steady state exergy balance gives:

$$\frac{q}{\dot{m}} \left[1 - \frac{T_0}{T_\sigma} \right] + a_{f1} = \frac{W}{\dot{m}} + a_{f2} + \frac{\dot{i}}{\dot{m}}$$



If there is no heat loss,

$$a_{f1} - a_{f2} = \frac{W}{\dot{m}} + \frac{\dot{i}}{\dot{m}}$$

the second law efficiency,

$$\eta_{II} = \frac{W/\dot{m}}{a_{f1} - a_{f2}}$$

2. COMPRESSOR AND PUMP:

Similarly for a compressor or pump,

$$\frac{W}{\dot{m}} = a_{f1} - a_{f2} + \frac{\dot{i}}{\dot{m}}$$

And

$$\eta_{II} = \frac{a_{f2} - a_{f1}}{-W/\dot{m}}$$

PROBLEMS:

Calculate the decrease in available energy when 25 kg of water at 95°C mix with 35 kg of water at 35°C, the pressure being taken as constant and the temperature of the surroundings being 15°C (c_p of water = 4.2 kJ/kg K).

Solution The available energy of a system of mass m , specific heat c_p , and at temperature T , is given by

$$A.E. = mc_p \int_{T_0}^T \left(1 - \frac{T_0}{T}\right) dT$$

∴ (A.E.)₂₅ = Available energy of 25 kg of water at 95°C

$$\begin{aligned} &= 25 \times 4.2 \int_{273+15}^{273+95} \left(1 - \frac{288}{T}\right) dT \\ &= 105 \left[(368 - 288) - 288 \ln \frac{368}{288} \right] \\ &= 987.49 \text{ kJ} \end{aligned}$$

(A.E.)₃₅ = Available energy of 35 kg of water at 35°C

$$\begin{aligned} &= 147 \left[(308 - 288) - 288 \ln \frac{308}{288} \right] \\ &= 97.59 \text{ kJ} \end{aligned}$$

Total available energy

$$\begin{aligned} (A.E.)_{\text{total}} &= (A.E.)_{25} + (A.E.)_{35} \\ &= 987.49 + 97.59 \\ &= 1085.08 \text{ kJ} \end{aligned}$$

After mixing, if t is the final temperature

$$25 \times 4.2 (95 - t) = 35 \times 4.2 (t - 35)$$

$$\begin{aligned} \therefore t &= \frac{25 \times 95 + 35 \times 35}{25 + 35} \\ &= 60^\circ\text{C} \end{aligned}$$

Total mass after mixing = 25 + 35 = 60 kg

(A.E.)₆₀ = Available energy of 60 kg of water at 60°C

$$\begin{aligned} &= 4.2 \times 60 \left[(333 - 288) - 288 \ln \frac{333}{288} \right] \\ &= 803.27 \text{ kJ} \end{aligned}$$

∴ Decrease in available energy due to mixing

$$\begin{aligned} &= \text{Total available energy before mixing} \\ &\quad - \text{Total available energy after mixing} \\ &= 1085.08 - 803.27 \\ &= 281.81 \text{ kJ} \end{aligned}$$

Ans.

2. The moment of inertia of a flywheel is $0.54 \text{ kg}\cdot\text{m}^2$ and it rotates at a speed 3000 RPM in a large heat insulated system, the temperature of which is 15°C . If the kinetic energy of the flywheel is dissipated as frictional heat at the shaft bearings which have a water equivalent of 2 kg, find the rise in the temperature of the bearings when the flywheel has come to rest. Calculate the greatest possible amount of this heat which may be returned to the flywheel as high-grade energy, showing how much of the original kinetic energy is now unavailable. What would be the final RPM of the flywheel, if it is set in motion with this available energy?

Solution Initial angular velocity of the flywheel

$$\omega_1 = \frac{2\pi N_1}{60} = \frac{2\pi \times 3000}{60} = 314.2 \text{ rad/s}$$

Initial available energy of the flywheel

$$\begin{aligned} &= (\text{K.E.})_{\text{initial}} = \frac{1}{2} I \omega_1^2 \\ &= 0.54 \text{ kg m}^2 \times (314.2)^2 \frac{\text{rad}^2}{\text{s}^2} \\ &= 2.66 \times 10^4 \text{ Nm} = 26.6 \text{ kJ} \end{aligned}$$

When this K.E. is dissipated as frictional heat, if Δt is the temperature rise of the bearings, we have

water equivalent of the bearings $\times$ rise in temperature = 26.6 kJ

$$\therefore \Delta t = \frac{26.6}{2 \times 4.187} = 3.19^\circ\text{C} \quad \text{Ans.}$$

$\therefore$ Final temperature of the bearings

$$t_f = 15 + 3.19 = 18.19^\circ\text{C}$$

The maximum amount of energy which may be returned to the flywheel as high-grade energy is

$$\begin{aligned} \text{A.E.} &= 2 \times 4.187 \int_{288}^{291.19} \left(1 - \frac{288}{T}\right) dT \\ &= 2 \times 4.187 \left[(291.19 - 288) - 288 \ln \frac{291.19}{288} \right] \\ &= 0.1459 \text{ kJ} \end{aligned}$$

The amount of energy rendered unavailable is

$$\begin{aligned} \text{U.E.} &= (\text{A.E.})_{\text{initial}} - (\text{A.E.})_{\text{returnable as high energy}} \\ &= 26.6 - 0.1459 \\ &= 26.4541 \text{ kJ} \end{aligned}$$

Since the amount of energy returnable to the flywheel is 0.146 kJ, if ω_2 is the final angular velocity, and the flywheel is set in motion with this energy

$$0.146 \times 10^3 = \frac{1}{2} \times 0.54 \omega_2^2$$

$$\therefore \omega_2^2 = \frac{146}{0.27} = 540.8$$

$$\therefore \omega_2 = 23.246 \text{ rad/s}$$

If N_2 is the final RPM of the flywheel

$$\omega_2 = 23.246 = \frac{2\pi N_2}{60}$$

$$\text{or } N_2 = \frac{23.246 \times 60}{2 \times \pi} = 222 \text{ RPM} \quad \text{Ans.}$$

3. Two kg of air at 500 kPa, 80°C expands adiabatically in a closed system until its volume is doubled and its temperature becomes equal to that of the surroundings which is at 100 kPa, 5°C. For this process, determine (a) the maximum work, (b) the change in availability, and (c) the irreversibility. For air, take $c_v = 0.718 \text{ kJ/kg K}$, $u = c_v T$ where c_v is constant, and $pV = mRT$ where p is pressure in kPa, V volume in m^3 , m mass in kg, R a constant equal to 0.287 kJ/kg K , and T temperature in K.

Solution From the property relation

$$TdS = dU + pdV$$

the entropy change of air between the initial and final states is

$$\int_1^2 dS = \int_1^2 \frac{mc_v dT}{T} + \int_1^2 \frac{mR dV}{V}$$

or
$$S_2 - S_1 = mc_v \ln \frac{T_2}{T_1} + mR \ln \frac{V_2}{V_1}$$

From Eq.

$$\begin{aligned} W_{\max} &= (U_1 - U_2) - T_0 (S_1 - S_2) \\ &= m \left[c_v (T_1 - T_2) + T_0 \left(c_v \ln \frac{T_2}{T_1} + R \ln \frac{V_2}{V_1} \right) \right] \\ &= 2 \left[0.718(80 - 5) + 278 \left(0.718 \ln \frac{278}{353} + 0.287 \ln \frac{2}{1} \right) \right] \\ &= 2 [53.85 + 278 (-0.172 + 0.199)] \\ &= 2 (53.85 + 7.51) \\ &= 122.72 \text{ kJ} \end{aligned}$$

Ans. (a)

From Eq the change in availability

$$\begin{aligned} &= \phi_1 - \phi_2 \\ &= (U_1 - U_2) - T_0 (S_1 - S_2) + p_0 (V_1 - V_2) \\ &= W_{\max} + p_0 (V_1 - V_2) \\ &= 122.72 + p_0 (V_1 - 2V_1) \\ &= 122.72 - 100 \times \frac{2 \times 0.287 \times 353}{500} \\ &= 82.2 \text{ kJ} \end{aligned}$$

Ans. (b)

The irreversibility

$$I = W_{\max, \text{useful}} - W_{\text{act}}$$

From the first law,

$$\begin{aligned} W_{\text{act}} &= Q - \Delta U = -\Delta U = U_1 - U_2 \\ \therefore I &= U_1 - U_2 - T_0 (S_1 - S_2) - U_1 + U_2 \\ &= T_0 (S_2 - S_1) \\ &= T_0 (\Delta S)_{\text{system}} \end{aligned}$$

For adiabatic process, $(\Delta S)_{\text{surr}} = 0$

$$\begin{aligned} I &= T_0 \left[mc_v \ln \frac{T_2}{T_1} + mR \ln \frac{V_2}{V_1} \right] \\ &= 278 \times 2 \left[0.718 \ln \frac{278}{353} + 0.287 \ln 2 \right] \\ &= 278 \times 2 (-0.172 + 0.199) \\ &= 15.2 \text{ kJ} \end{aligned}$$

Ans. (c)

4. Air expands through a turbine from 500 kPa, 520°C to 100 kPa, 300°C. During expansion 10 kJ/kg of heat is lost to the surroundings which is at 98 kPa, 20°C. Neglecting the K.E. and P.E. changes, determine per kg of air (a) the decrease in availability, (b) the maximum work, and (c) the irreversibility.

For air, take $c_p = 1.005 \text{ kJ/kg K}$, $h = c_p T$ where c_p is constant, and the p , V and T relation

Solution From the property relation

$$TdS = dH - V dp$$

the entropy change of air in the expansion process is

$$\int_1^2 dS = \int_1^2 \frac{m c_p dT}{T} - \int_1^2 \frac{m R dp}{p}$$

or
$$S_2 - S_1 = m c_p \ln \frac{T_2}{T_1} - m R \ln \frac{p_2}{p_1}$$

For 1 kg of air,

$$s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{p_2}{p_1}$$

From Eq. the change in availability

$$\begin{aligned} \psi_1 - \psi_2 &= b_1 - b_2 \\ &= (h_1 - T_0 s_1) - (h_2 - T_0 s_2) \\ &= (h_1 - h_2) - T_0 (s_1 - s_2) \\ &= c_p (T_1 - T_2) - T_0 \left(R \ln \frac{p_2}{p_1} - c_p \ln \frac{T_2}{T_1} \right) \\ &= 1.005 (520 - 300) - 293 \left(0.287 \ln \frac{1}{5} - 1.005 \ln \frac{573}{793} \right) \\ &= 1.005 \times 220 - 293 (0.3267 - 0.4619) \\ &= 221.1 + 39.6 \\ &= 260.7 \text{ kJ/kg} \end{aligned}$$

Ans. (a)

The maximum work is

$$\begin{aligned} W_{\max} &= \text{change in availability} = \psi_1 - \psi_2 \\ &= 260.7 \text{ kJ/kg} \end{aligned}$$

Ans. (b)

From S.F.E.E.,

$$\begin{aligned} Q + h_1 &= W + h_2 \\ W &= (h_1 - h_2) + Q \\ &= c_p (T_1 - T_2) + Q \\ &= 1.005 (520 - 300) - 10 \\ &= 211.1 \text{ kJ/kg} \end{aligned}$$

The irreversibility

$$\begin{aligned} I &= W_{\max} - W \\ &= 260.7 - 211.1 \\ &= 49.6 \text{ kJ/kg} \end{aligned}$$

Ans. (c)

Alternatively,

$$\begin{aligned} I &= T_0 (\Delta S_{\text{system}} + \Delta S_{\text{surr}}) \\ &= 293 \left[1.005 \ln \frac{573}{793} - 0.287 \ln \frac{1}{5} + \frac{10}{293} \right] \\ &= 293 \times 0.1352 + 10 \\ &= 49.6 \text{ kJ/kg} \end{aligned}$$

Ans. (d)

5. An air preheater is used to cool the products of combustion from a furnace while heating the air to be used for combustion. The rate of flow of products is 12.5 kg/s and the products are cooled from 300 to 200°C, and for the products at this temperature $c_p = 1.09$ kJ/kg K. The rate of air flow is 11.5 kg/s, the initial air temperature is 40°C, and for the air $c_p = 1.005$ kJ/kg K. (a) Estimate the initial and final availability of the products. (b) What is the irreversibility for the process? (c) If the heat transfer from the products occurs reversibly through heat engines, what is the final temperature of the air? What is the power developed by the heat engine? Take $T_0 = 300$ K and neglect pressure drop for both the fluids and heat transfer to the surroundings.

Solution

(a) ψ_1 = initial availability of the products

$$= (h_1 - h_0) - T_0(s_1 - s_0)$$

$$= c_{p_g}(T_{g1} - T_0) - T_0 c_{p_g} \ln \frac{T_{g1}}{T_0}$$

$$= 1.09(573 - 300) - 300 \times 1.09 \ln \frac{573}{300}$$

$$= 297.57 - 211.6 = 39.68 \text{ kJ/kg}$$

ψ_2 = final availability of the products

$$= (h_2 - h_0) - T_0(s_2 - s_0)$$

$$= 1.09(473 - 300) - 300 \times 1.09 \ln \frac{473}{300}$$

$$= 188.57 - 148.89 = 39.68 \text{ kJ/kg}$$

(b) Decrease in availability of the products

$$= \psi_1 - \psi_2$$

$$= (h_1 - h_2) - T_0(s_1 - s_2)$$

$$= 1.09(573 - 473) - 300 \times 1.09 \ln \frac{573}{473}$$

$$= 109 - 62.72 = 46.28 \text{ kJ/kg}$$

By making an energy balance for the air preheater

$$\dot{m}_g c_{p_g}(T_{g1} - T_{g2}) = \dot{m}_a c_{p_a}(T_{a2} - T_{a1})$$

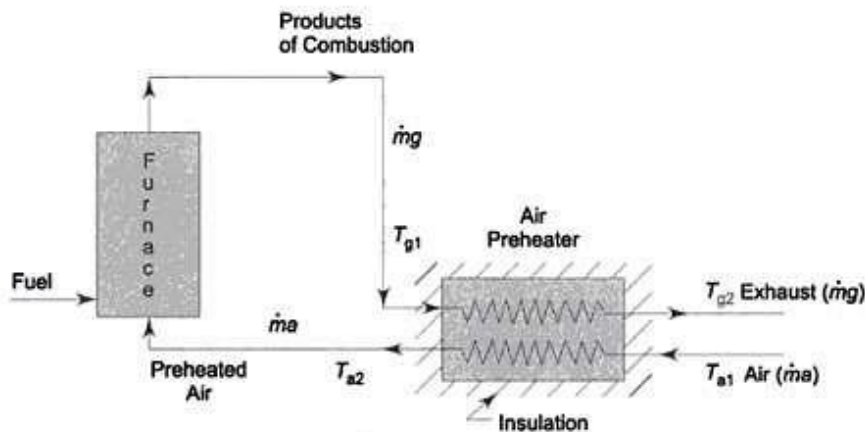
$$12.5 \times 1.09(573 - 473) = 11.5 \times 1.005(T_{a2} - 313)$$

$$\therefore T_{a2} = \frac{12.5 \times 109}{11.5 \times 1.005} + 313 = 430.89 \text{ K}$$

Increase in availability for air

$$= \psi_2 - \psi_1$$

$$= (h_2 - h_1) - T_0(s_2 - s_1)$$

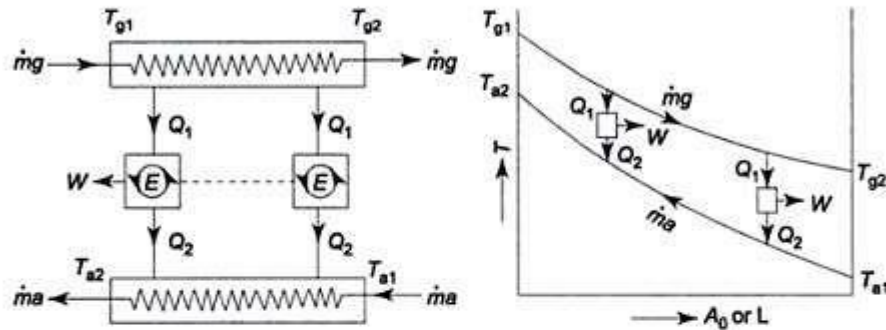


$$\begin{aligned}
 &= c_{p_a} (T_{a2} - T_{a1}) - T_0 c_{p_a} \ln \frac{T_{a2}}{T_{a1}} \\
 &= 1.005 \times (430.89 - 313) - 300 \times 1.005 \ln \frac{430.89}{313} \\
 &= 118.48 - 96.37 = 22.11 \text{ kJ/kg}
 \end{aligned}$$

∴ Irreversibility of the process

$$\begin{aligned}
 &= 12.5 \times 46.28 - 11.5 \times 22.11 \\
 &= 578.50 - 254.27 \\
 &= 324.23 \text{ kW}
 \end{aligned}$$

(c) Let us assume that heat transfer from the products to air occurred through heat engines reversibly as shown in Fig.



For reversible heat transfer,

$$\Delta \dot{S}_{\text{univ}} = 0$$

$$\Delta \dot{S}_{\text{sys}} + \Delta \dot{S}_{\text{surr}} = 0$$

$$\Delta \dot{S}_{\text{gas}} + \Delta \dot{S}_{\text{air}} = 0$$

$$\Delta \dot{S}_{\text{gas}} = -\Delta \dot{S}_{\text{air}}$$

$$\dot{m}_g c_{p_g} \ln \frac{T_{g2}}{T_{g1}} = -\dot{m}_a c_{p_a} \ln \frac{T_{a2}}{T_{a1}}$$

$$12.5 \times 1.09 \ln \frac{473}{573} = -11.5 \times 1.005 \ln \frac{T_{a2}}{313}$$

$$\therefore T_{a2} = 392.41 \text{ K}$$

Rate of heat supply from the gas to the working fluid in the heat engine,

$$\begin{aligned}
 \dot{Q}_1 &= \dot{m}_g c_{p_g} (T_{g1} - T_{g2}) \\
 &= 12.5 \times 1.09 (573 - 473) \\
 &= 1362.50 \text{ kW}
 \end{aligned}$$

Rate of heat rejection from the working fluid in the heat engine to the air,

$$\begin{aligned}
 \dot{Q}_2 &= \dot{m}_a c_{p_a} (T_{a2} - T_{a1}) \\
 &= 11.5 \times 1.005 (392.41 - 313) \\
 &= 917.78 \text{ kW}
 \end{aligned}$$

Total power developed by the heat engine

$$\begin{aligned}
 \dot{W} &= \dot{Q}_1 - \dot{Q}_2 = 1362.50 - 917.78 \\
 &= 444.72 \text{ kW}
 \end{aligned}$$